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MASTER'S STUDY PROPOSAL

隱形帳簿：量化印尼平台經濟中的隱形缺口
The Invisible Ledger: Quantifying the Invisible Wedge in Indonesia's Platform Economy

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Abstract

The digital economy is measured through several indicators - platform transaction value, platform revenue, national e-commerce statistics and payment data - that capture different economic objects but are often used as if they were interchangeable. This study develops a task-specific framework for asking when an observable indicator is an adequate proxy for the economic activity it is used to represent. I retain the invisible wedge as transaction value minus recognized revenue, use the Ecosystem Ratio to describe the level relationship, and use annual growth divergence to test whether that relationship remains stable through time. In the issuer-reported Indonesian sample, revenue grows faster than transaction value in six of eight annual comparisons, the two measures move in opposite directions twice, and the median absolute divergence is 38.92 percentage points. Tokopedia is the clearest mechanism case: transaction value fell 8.9 percent in 2022-2023 while third-party net revenue rose 53.2 percent; 60.6 percent of that increase reconciles to a lower customer-incentive deduction and 39.4 percent to higher gross revenue. BPS-Statistics Indonesia reports total e-commerce value growing 17.08 percent in 2023-2024 while its marketplace category grew 1.45 percent. The contribution is therefore not one universally correct measure of the digital economy, but a framework for testing when indicator choice changes conclusions about scale, growth, or direction.

1. Introduction

Southeast Asia's digital economy grew from approximately US\$100 billion in 2020 to US\$263 billion in 2024 (Google, Temasek and Bain 2020, 2024). That headline is one indicator of the activity: the gross value of transactions involved. Digital platforms report another number, their own recognized revenue; national statistical agencies report online sales by surveyed businesses; and payment systems report funds moved. Each series is valid for the object it was designed to measure, but the indicators are not interchangeable. The distinction matters because investors, analysts and public institutions routinely use observable records as stand-ins for broader economic activity. If the relationship between the record and the intended target changes, the measured scale, growth and even direction of activity can change with the indicator selected.

Suppose a platform processes 100 units of transaction value but recognizes only 10 as its own revenue: the commissions and fees it earns for arranging the transactions. Depending on the business model, the remaining 90 can include merchant receipts, driver payouts, inventory costs, taxes and other pass-through payments. An analyst using the platform's accounts sees 10, while one measuring the commerce it processes sees 100. I call the difference the invisible wedge (W). The term refers to what a revenue-only measure excludes, not to unrecorded activity. Revenue records what the platform earned, while transaction value records what it carried. The arithmetic is familiar; the measurement problem is that the relationship between the two does not stay fixed.

Existing research explains why the value of commerce processed through a platform can far exceed the revenue the platform records (Rochet and Tirole 2003; Hagiu and Wright 2015; Evans and Schmalensee 2016), and why revenue-recognition rules affect how much of that difference appears in the accounts (International Accounting Standards Board 2014). What is less clear is whether the relationship is stable enough for one observable to stand in for another. A large but stable level gap may still preserve the same growth story. A changing relationship can instead make revenue and transaction value imply different growth rates or even opposite directions. Monetization, customer incentives, business mix, principal-agent treatment and reporting scope can all move that relationship. The empirical problem is therefore not to choose one universally correct record, but to determine whether a specified indicator is sufficient for a specified economic inference, scope and period.

Against this backdrop, I study Indonesia, the largest digital market in Southeast Asia. Platform reports disclose both the commerce a platform carries and the revenue it recognizes, so the consequences of substituting one for the other can be observed directly within the same accounts. The study covers four marketplaces, including Tokopedia, Blibli, Bukalapak and Shopee, and Grab, whose Indonesian business is mainly delivery and other on-demand services. Tokopedia, Blibli and Bukalapak report both measures themselves (i.e., issuer-reported figures). For Grab and Shopee, Indonesian figures must be constructed from company-wide ratios and a third-party estimate.¹ To limit the concerns this raises, I follow each platform against itself through time and keep issuer-reported and constructed figures apart, so that the company, the business activity and the reporting convention stay as constant as the disclosures allow. I then compare the platform evidence with two independent indicators: national e-commerce statistics from BPS-Statistics Indonesia (BPS), the national statistical agency, and payment statistics from Bank Indonesia, the central bank. Only platform transaction value and revenue enter the wedge; BPS and payment data serve as comparison indicators. Because their definitions and scopes differ, they are not yet directly comparable with platform figures, and each answers a different question.

¹ See Section 4.3 for more details.

In 2023, Tokopedia, Shopee and Grab together processed US\$43.23 billion of transaction value against US\$3.16 billion of recognized revenue. The level gap is large, but the more consequential test is whether revenue preserves the growth story of the commerce underneath it. Tokopedia shows that it may not: between 2022 and 2023, third-party net revenue rose 53.2 percent while transaction value fell 8.9 percent. Of the net-revenue increase, 60.6 percent reconciles to a lower customer-incentive deduction and 39.4 percent to higher gross revenue. Among the three platforms reporting both figures themselves, revenue grew faster in six of eight annual comparisons and the measures moved in opposite directions twice. BPS shows a related national measurement problem: total e-commerce value grew 17.08 percent in 2023-2024 while its marketplace category grew 1.45 percent, so 98.46 percent of the measured increase occurred outside that category. Across eight listed platforms outside Indonesia, revenue grew faster in 22 of 29 clean comparisons, but the median absolute divergence was 7.1 percentage points compared with 38.92 in the Indonesian issuer-reported sample. The issue is therefore not which indicator is false, but when one can stand in for another without changing the economic conclusion.

A statement about the size or growth of Indonesia's digital economy is therefore also a statement about the indicator used. Revenue describes recognized platform performance; transaction value describes commerce carried; BPS marketplace and total e-commerce measures answer different channel questions. Substitution is defensible only when the relationship is adequate for the intended task. That matters to investors, statistical measurement and institutions that act on platform-linked records, and it motivates a later finance test of which indicator or revenue-growth component is most informative for stock returns.

1.1 Research question and objectives

One question drives the proposal:

When can platform revenue serve as a useful proxy for the commerce a platform carries, and when does choosing one indicator rather than another change conclusions about scale, growth or direction?

Indonesia provides the main empirical setting. Three objectives follow. First, I measure the level and movement of the transaction-revenue relationship from 2020 to 2025 using W, E and D. Second, I reconcile selected large divergences with disclosed changes in customer incentives, monetization, revenue recognition and reporting scope. Third, I compare the platform evidence with national e-commerce, payment and external-platform indicators to determine when the records support the same economic conclusion and when indicator choice changes it.

A separate comparison of eight listed platform businesses outside Indonesia provides an external benchmark for whether the same separation recurs under other business models and periods.

1.2 Contributions

My study contributes in four ways. First, it turns the transaction-revenue wedge from a one-year scale comparison into a task-specific measurement problem. W and E describe the level relationship, while D tests whether it remains stable through time. In the issuer-reported sample, revenue outgrew transaction value in six of eight comparisons, with two sign reversals and a median absolute divergence of 38.92 percentage points.

Second, I reconcile a major divergence using disclosed accounting components. In 2022-2023, Tokopedia's transaction value fell 8.9 percent while third-party net revenue rose 53.2 percent; 60.6 percent of the net-revenue increase reconciles to a lower customer-incentive deduction and 39.4 percent to higher gross revenue. The reconciliation is arithmetic rather than causal, but it identifies an observable source of the divergence.

Third, I extend the comparison beyond one company and one measurement system. Across eight listed platforms outside Indonesia, revenue grew faster in 22 of 29 clean comparisons but the median absolute divergence was only 7.1 percentage points. BPS total e-commerce and its marketplace category also grew at materially different rates in 2023-2024. Proxy adequacy therefore depends on the relationship, scope and period.

Finally, I address a measurement problem in finance while keeping evidence tiers explicit. Issuer-reported pairs are kept separate from constructed country measures, while national statistics and payment data remain comparison indicators rather than being forced into one total. The framework asks which observable is sufficient for a stated inference without claiming that one record is universally superior, and it leads directly to the advisor's proposed stock-return extension.

2. Literature Review

Digital platforms sit between economic activity and the indicators used to measure it. They facilitate transactions between buyers and sellers, recognize only part of that activity as their own revenue, and generate records that can also appear in payment, statistical and administrative systems. Existing research explains each part of this process from a different angle. Taken together, it provides the foundation for asking not only why the records differ, but when one observable record can serve as a useful stand-in for another economic target.

2.1 Platform Economics and Revenue Recognition

Platform economics explains why the value of commerce processed through a platform can be much larger than the revenue the platform records. Digital platforms often connect consumers with merchants or other service providers and earn commissions, fees, advertising, logistics, or other intermediation revenue rather than the full value of every transaction they facilitate (Rochet and Tirole 2003; Caillaud and Jullien 2003; Parker and Van Alstyne 2005; Armstrong 2006; Hagiu and Wright 2015; Evans and Schmalensee 2016).

In particular, revenue-recognition rules affect how much of this difference appears in the accounts. Under International Financial Reporting Standard 15 (IFRS 15), whether a company acts as a principal or an agent affects whether it recognizes the full customer payment or only the amount it earns from arranging the transaction (International Accounting Standards Board 2014). Customer incentives, monetization, service mix, acquisitions, and changes in business scope can also move reported revenue without an equivalent change in transaction activity. De Franco, Kothari and Verdi (2011) show why comparable accounting bases matter when financial information is interpreted. Overall, the literature explains why a transaction–revenue gap can exist and move over time, but not how economically large that gap becomes in practice.

2.2 Measuring the Digital Economy

Official measurement frameworks already distinguish between the commerce facilitated through a platform and the service provided by the platform itself. National-accounting guidance separates the buyer-seller transaction from the platform's intermediation service rather than treating the full transaction value as platform output (Ahmad and Schreyer 2016; International Monetary Fund 2018; OECD 2023; United Nations et al. 2025).

In practice, the quantities are published by different producers. For instance, industry reports size the regional digital economy by gross merchandise value (Google, Temasek and Bain 2024), national statistics measure the online sales of surveyed businesses (BPS-Statistics Indonesia 2025), and listed platforms report the revenue they recognize under IFRS 15 (International Accounting Standards Board 2014). A difference that is harmless when the measures move together becomes important when they do not, because the choice of measure can change conclusions about economic scale and

growth. Overall, this study compares the indicators rather than assuming that one can substitute for another.

2.3 Digital Records and Third-Party Information

Transactions that do not enter a platform's revenue can still leave detailed records within the platform's system. Merchants and service providers can be small, self-employed, or weakly represented in conventional financial records, yet the transactions they make on a platform are recorded by it. This differs from shadow-economy settings, where reliable formal records may be sparse (La Porta and Shleifer 2014; Ulyssea 2018; Medina and Schneider 2019). Platform-mediated activity can also create traces outside conventional records (Berg et al. 2020; Barrios, Hochberg and Yi 2022; Denes, Lagaras and Tsoutsoura 2025).

Third-party information changes compliance and enforcement (Kleven et al. 2011; Pomeranz 2015; Naritomi 2019; Kleven, Kreiner and Saez 2016; Slemrod 2019). The OECD Model Rules (OECD 2020) and the European Union's DAC7 regime (European Union 2021) apply this principle to platforms, as does Indonesia's Minister of Finance Regulation No. 37 of 2025 (PMK 37/2025) through seller identification, transaction-linked reporting and marketplace withholding (Ministry of Finance of the Republic of Indonesia 2025). Overall, the relevance here is informational: what a measure or rule can show depends on the records it uses.

2.4 Research Gap

Existing research explains the pieces of the problem separately. Platform economics explains why transaction value can exceed revenue; accounting explains why the gap can move; official measurement frameworks explain why these quantities should not be treated as equivalent; and research on digital records explains why some platform-mediated activity outside revenue can remain observable in platform records.

However, the remaining research gap is not simply that the published numbers differ. What is less established is whether a candidate indicator is sufficient for the inference being made: how far it sits from the intended target in level, whether it preserves growth and direction, and what disclosed components explain important divergences. This study addresses that gap by following the transaction-revenue relationship through time, reconciling selected movements, and using national and external evidence to test the boundary of the inference.

3. Theoretical Framework and Hypotheses

Digital platform activity can be observed through several indicators, but they do not measure the same economic object. The framework is task-specific: define the target, match an observable with the same period and scope, separate the level relationship from the dynamic relationship, reconcile important divergences, and state the inference only for the setting the evidence supports.

For the platform analysis, transaction value and revenue are paired only when they refer to the same period and business activity. This scope match is the first validity condition. Issuer-reported and constructed figures remain separate, as do national statistics and payment indicators; the study tests relationships between records rather than pooling unlike measures.

3.1 Measurement Framework

Let V_{it} denote transaction value processed by platform i in period t , and R_{it} the revenue recognized over the same matched period and business scope. The absolute “invisible wedge” is:

$$W_{it} = V_{it} - R_{it}$$

W is transaction value outside the platform's recognized revenue, not an omission from its accounts: both inputs can appear in the same company report. It is a transaction-value measure rather than an

income or value-added measure; Section 7 states the corresponding inference boundaries. Because platforms differ in size, I also express the wedge relative to revenue. The “Ecosystem Ratio” is:

$$E_{it} = (V_{it} - R_{it}) / R_{it}$$

An E of 9 means that nine units of transaction value lie outside the revenue line for every unit recognized as platform revenue. E contains the same information as the platform's monetization rate, $t = R/V$, because $E = 1/t - 1$. For example, a platform that recognizes 2.5 percent of transaction value as revenue has an E of 39. The residual form matches the research question, which concerns commerce outside recognized revenue.

The Ecosystem Ratio is constructed for this study as a descriptive transformation of the monetization rate. Its interpretation rests on the platform-economics result that booked revenue need not track the transaction value a multisided platform coordinates (Rochet and Tirole 2003; Hagiu and Wright 2015; Evans and Schmalensee 2016), together with the accounting result that the share entering revenue depends on how revenue is recognized, including whether the platform acts as principal or agent (International Accounting Standards Board 2014; De Franco, Kothari and Verdi 2011).

A single ratio does not show whether the relationship holds through time. W and E therefore answer a level question, while annual growth divergence tests dynamic sufficiency. Where $g(X)$ is the percentage growth of X from one year to the next, I define:

$$D_{it} = g(R_{it}) - g(V_{it})$$

A positive D means revenue grew faster than transaction value; a negative D means transaction value grew faster. If revenue were a stable dynamic proxy, D would remain close to zero over the relevant window. A sign reversal, where one measure rises and the other falls, is the strongest observable failure because indicator choice changes the direction of the conclusion. W gives the wedge in currency units, E describes the relative level relationship, and D carries the central dynamic test. The framework does not presume failure: a relationship that remains close in level and movement can be adequate for a stated task.

3.2 Hypothesis Development

The four hypotheses below test the research question one indicator at a time. H1 concerns transaction value and revenue within platforms, H2 and H3 the national e-commerce evidence, and H4 the comparison across indicators. None is a causal claim. Table 1 pairs each hypothesis with the evidence that tests it.

Platform economics and accounting both suggest that the share of transaction value a platform recognizes as revenue is not fixed. Revenue depends on monetization, customer incentives and whether the platform acts as principal or agent under IFRS 15, and these can change with the platform's business model, scope and pricing (Rochet and Tirole 2003; Hagiu and Wright 2015; International Accounting Standards Board 2014). If the relationship were stable, revenue would be a workable stand-in for the commerce a platform carries, D would remain close to zero and E would be constant. I therefore hypothesize that:

H1. Within the same platform, transaction value and platform revenue do not grow in proportion.

The test is whether D remains close to zero; changes in its sign identify especially strong reversals but are not required for H1.

National e-commerce value can grow because existing businesses sell more or because more businesses sell online. Growth driven by new sellers can take place in channels that platform accounts do not cover, such as social media and instant messaging, which BPS reports are widely used (BPS-Statistics Indonesia 2025). Because BPS reports both the number of e-commerce

businesses and total e-commerce value, national growth can be divided arithmetically into these two components (Section 4.4). I therefore hypothesize that:

H2. National e-commerce growth comes mainly from more participating businesses rather than from higher sales per business.

The split is arithmetic and does not assign new businesses to particular channels.

Transactions made on a platform are recorded by the platform, and marketplace participation may accompany more formal recordkeeping by the sellers themselves (Section 2.3). Financial statements are one conventional record a business keeps of its own activity. I therefore hypothesize that:

H3. Marketplace participation is associated with stronger financial recordkeeping.

H3 tests whether the prevalence of financial statements differs between marketplace and non-marketplace sellers; it does not treat the association as causal.

If H1 holds, revenue growth and transaction-value growth can support different conclusions about the same platform. If most national e-commerce growth occurs outside the marketplace channel, marketplace figures can likewise support different conclusions from figures for all e-commerce. H4 combines these two comparisons. I therefore hypothesize that:

H4. Conclusions about the growth of Indonesian digital commerce change with the indicator used to measure it.

H4 is the central claim of the study. It fails if revenue grows in step with transaction value within platforms and the marketplace channel grows in step with all e-commerce. Level differences between platform and BPS figures, and payment growth, are reported as context because their definitions and coverage differ.

To summarize, H1 asks whether revenue is a stable guide to transaction value within platforms; H2 and H3 describe where national e-commerce growth comes from and how marketplace participation relates to recordkeeping; and H4 asks whether the choice of indicator changes the conclusion.

Table 1. Hypotheses and principal empirical tests

This table pairs each hypothesis with the principal empirical test used to examine it. H1 concerns transaction value and revenue within platforms, H2 and H3 the national e-commerce evidence, and H4 the comparison across indicators. D denotes annual growth divergence and E the Ecosystem Ratio, both defined in Section 3.1. BPS denotes BPS-Statistics Indonesia. None of the tests is interpreted as a causal claim.

Hypothesis	Principal empirical test
H1 - Transaction-revenue divergence	Growth divergence D and Ecosystem Ratio E for issuer-reported pairs: level relationship, dynamic stability and sign reversals
H2 - Sources of e-commerce growth	BPS national e-commerce value split into business count and sales per business
H3 - Marketplace participation and recordkeeping	BPS published comparison of marketplace and non-marketplace sellers
H4 - Measurement choice	Growth of revenue against transaction value within platforms, and of the marketplace channel against all e-commerce in BPS

The comparison of platform businesses outside Indonesia corroborates H1 rather than forming a fifth hypothesis.

4. Data and Methodology

The study uses several indicators because no single one captures every part of digital platform activity. Platform reports provide transaction value and revenue; BPS-Statistics Indonesia provides national e-commerce and business evidence; Bank Indonesia provides payment statistics; and platform companies outside Indonesia provide a separate comparison. The indicators answer different questions and are not combined into one dataset.

4.1 Selecting the Platform Evidence

The basic observation is one platform in one year, with transaction value and revenue referring to the same period and, as closely as possible, the same business activity. An observation is retained only when its period, geographic and business coverage, units, definitions, and source can be identified; measures referring to different periods are excluded, and major changes in accounting definitions or business structure are recorded.

The main sample consists of the platforms that report transaction value and revenue themselves: Tokopedia, Bilibli and Bukalapak. Grab and Shopee, whose Indonesian figures must be constructed, support the 2023 scale comparison. Lazada and TikTok Shop, both large in Indonesia, are outside the sample because no matched Indonesian revenue series has been located for either. The platforms are disclosing cases rather than a representative sample of Indonesian platforms.

Table 2. Indonesian platform evidence

This table reports the Indonesian platform evidence used in the analysis. Issuer-reported figures are disclosed by the platform; figures that require construction are derived from company-wide ratios or a third-party estimate (Table 3). The number of platform-year observations is given in parentheses, and each series covers only years for which a matched pair exists. The main sample comprises the issuer-reported rows (11 platform-years); adding Grab and Shopee gives 17.

Type of evidence	Platforms	Years (observation)	Role
Issuer-reported, Indonesia-aligned	Tokopedia e-commerce	2022–2023 (2)	Main sample: cleanest scope, shortest window
Issuer-reported, broader scope	Bilibli third-party business; Bukalapak Group	Bilibli 2021–2025; Bukalapak 2020–2023 (9)	Main sample: most of the time series
Requires construction	Grab; Shopee Indonesia	Grab 2021–2023; Shopee 2022–2024 (6)	Supporting: 2023 scale comparison

As shown in Table 2, the main sample has 11 platform-years and 8 year-to-year comparisons; adding Grab and Shopee raises these to 17 and 12. Each series covers only years for which a matched pair exists. Tokopedia's 2021 pair covers mismatched periods, and after Tokopedia was combined with TikTok Shop under PT Tokopedia on 31 January 2024, the listed parent reports a contractual fee rather than platform revenue. Bilibli's third-party segment includes online travel; its 2020 prospectus pair is used only as a sensitivity. Bukalapak reports at Group level, including some overseas activity; its 2024 pair is excluded because it covers nine months of transactions against twelve of revenue. Grab's transaction measure changes basis after 2023.

4.2 Comparing Platform Activity Through Time

Changes through time are examined mainly by comparing each platform with itself. This keeps the company, business activity, and reporting convention as similar as the disclosures allow. When transaction value and revenue move sharply apart, disclosed changes in monetization, customer incentives, revenue recognition, and business scope are used to reconcile the movement where the filings permit; these reconciliations are arithmetic rather than causal.

Companies use different names for transaction activity, including gross transaction value (GTV), gross merchandise value (GMV), and total payment volume (TPV). Transaction value, V, is the umbrella concept in this proposal and is labeled GTV; each issuer's own label and definition is retained and checked rather than assumed to be identical.²

Growth rates are computed in each issuer's reporting currency, so exchange-rate movements do not enter D. US dollar values are used only for the 2023 cross-section, where rupiah figures are converted at the World Bank's 2023 official exchange rate of 15,236.88 rupiah per dollar (World Bank n.d.).

4.3 Direct and Constructed Indonesia Measures

Not every company reports the same Indonesia-specific information. Table 3 sets out, for each platform, what is disclosed directly, what has to be constructed, and the limitation that follows.

Table 3. Construction of the principal platform measures

This table sets out, for each platform, the figures disclosed directly, the figures that must be constructed, and the main limitation that follows. Indonesia GTV for Grab is estimated as Indonesia revenue $\times$ Group GTV / Group revenue, and Indonesia revenue for Shopee as Indonesia transaction value $\times$ Group monetization rate (Section 4.3). The Shopee Indonesian transaction estimate is a third-party estimate from Momentum Works (2024).

Platform	Directly available	Constructed or external input	Main limitation
Tokopedia	E-commerce transaction value and segment revenue	None for matched pair	Indonesia-aligned segment, not literal country line
Grab	Indonesia revenue; Group revenue and GTV	Indonesia GTV derived from Group figures	Indonesian ratio equals the Group ratio
Shopee	Group service monetization rate	External Indonesia transaction estimate; Indonesia revenue derived	Indonesian ratio set by the Group rate
Blibli / Bukalapak	Reported transaction and revenue pairs	None for retained pairs	Scope extends beyond Indonesian goods commerce

Grab reports Indonesian revenue directly but not Indonesian transaction value. Indonesian GTV is therefore estimated as:

$$\text{Indonesia GTV} = \text{Indonesia revenue} \times \text{Group GTV} / \text{Group revenue}$$

The Group monetization rate is Group revenue divided by Group gross transaction value.

Sea Limited does not separately report Shopee's transaction value for Indonesia. Momentum Works supplies the Indonesian estimate; its annual Southeast Asian e-commerce report is widely cited for country-level platform estimates and is used because no issuer discloses the figure. The value is labeled as a third-party estimate wherever it appears, and Indonesian revenue is then estimated using Sea's Group service monetization rate:

$$\text{Indonesia revenue} = \text{Indonesia transaction value} \times \text{Group monetization rate}$$

Construction fixes both Indonesian ratios to company-wide relationships: Grab's E equals its Group E, while Shopee's equals the inverse of Sea's monetization rate minus one. The series contribute Indonesian levels but no independent evidence on the ratio or its movement, so they remain outside the main sample.

² Issuer labels differ. GoTo reports GTV and Sea Limited reports GMV for Shopee; for these the label differs but the measure is the same. Blibli and Bukalapak report TPV for the segments used here, and Grab reports GMV for deliveries and mobility and TPV for financial services. Where a source reports TPV the measure rests on a different basis, so it is retained under its own label rather than relabeled as GTV.

4.4 Evidence Beyond Platform Accounts

Table 4. Indicators used and their role

This table lists the indicators used and the role of each. The indicators answer different questions, so they are reported separately rather than pooled into one estimate. BPS denotes BPS-Statistics Indonesia. Appendix A provides variable definitions in greater detail.

Evidence source	What is observed	Role
Indonesian platform histories	Transaction value and revenue through time	Test the level relationship and dynamic stability of revenue against transaction value
BPS national e-commerce statistics	Online sales of surveyed businesses, by channel, and the number of businesses	Examine the level and sources of growth
BPS marketplace study	Business-level marketplace and recordkeeping evidence	Examine the association with financial recordkeeping
Bank Indonesia	Economy-wide digital payment measures	Context on payment growth
Platform companies outside Indonesia	Matched transaction and revenue measures	Test whether the same boundary appears elsewhere

As shown in Table 4, each indicator has a single role. The indicators answer different questions, so they are reported separately rather than pooled into one estimate.

For H2, national e-commerce value S is written as:

$$S = N \times A$$

where N is the estimated number of e-commerce businesses and A is implied nominal sales per business, and the change in S is divided arithmetically between the two. BPS publishes two counts of 2023 businesses; the count consistent with its stated growth figure is used, and the other is retained as a sensitivity check.

For H3, BPS's published business-level comparison of marketplace and non-marketplace sellers is used. For H4, BPS's split of e-commerce value by sales channel is set against the platform evidence. Bank Indonesia's economy-wide payment measures provide an independent view of digitalization. Because they cover more than e-commerce and also reflect adoption and substitution between payment methods, they are compared by growth rate and are not treated as sales estimates.

4.5 External Evidence and Robustness

The external comparison contains 48 matched annual observations for eight listed platform businesses from 2017 to 2025: eBay, Etsy, Jumia, Mercado Libre, Rakuten, Sea, Shopify and Zalando. It tests whether the same separation between transaction value and revenue appears under other business models and is reported separately from the Indonesian evidence.

Robustness checks vary the sample-inclusion rules, revenue definitions, publication versions, and the assumptions in the constructed country measures. The resulting ranges are reported as sensitivity ranges rather than statistical confidence intervals.

5. Preliminary Evidence and Feasibility

The evidence assembled so far establishes that the proposed comparisons can be carried out and that proxy adequacy is an empirical rather than definitional question. The results first show the scale and movement of the transaction-revenue relationship, then compare that relationship with national and external evidence. The objective is not to declare one record correct and another wrong, but to identify when a record supports the same economic inference as the target and when it does not.

5.1 Platform Scale and Movement

The 2023 comparison covers Shopee and Tokopedia, which together represented about 70 percent of estimated Indonesian marketplace transaction value, and Grab, a major delivery intermediary.

Table 5. Indonesia-focused 2023 platform comparison

This table reports the Indonesia-focused 2023 platform comparison, in US\$ billions. V is transaction value, R is recognized revenue, $W = V - R$ is the invisible wedge, and $E = (V - R) / R$ is the Ecosystem Ratio (Section 3.1). Rupiah figures are converted at the World Bank's 2023 official exchange rate of 15,236.88 rupiah per dollar. Grab's transaction value is constructed from Group figures, and Shopee's transaction value and revenue are estimated; the three-platform row is the sum of the three cases, not a national total. Appendix A provides variable definitions in greater detail.

Case	Transaction value (V)	Revenue (R)	Wedge (W)	Ecosystem Ratio (E)	Evidence
Grab Indonesia	5.381	0.605	4.776	7.895x	Transaction value constructed
Tokopedia	16.331	0.405	15.926	39.296x	Issuer-reported pair
Shopee Indonesia	21.520	2.152	19.368	9.000x	Transaction value and revenue estimated
Three platforms	43.233	3.162	40.070	12.671x	Sum of three cases, not national total

Table 5 reports that cross-section. Across the three platforms, US\$43.23 billion of transaction value corresponds to US\$3.16 billion of recognized revenue, a wedge of US\$40.07 billion and a transaction-value-to-revenue ratio of 13.7 to 1 ($E = 12.7$). This establishes a large level separation, but level separation alone does not determine whether revenue is a useful proxy for growth. That question requires the within-platform dynamic evidence reported below.

One concern is that two ratios rest on construction. Shopee's 9.0 follows from an assumed company-wide monetization rate of 10 percent, and Grab's 7.9 is its company-wide ratio, so neither provides independent Indonesian ratio evidence. To address this concern, I vary Shopee's rate from 9 to 11 percent, which puts the combined wedge between US\$39.86 billion and US\$40.29 billion, and exclude each platform in turn, which leaves a wedge of at least US\$20.70 billion.

At the 2023 exchange rate, Blibli's and Bukalapak's reported transaction and revenue figures yield wedges of US\$3.20 billion and US\$10.50 billion. They are excluded from the total because Blibli's measure includes online travel and Bukalapak's is reported at Group level. Their ratios, 43.4 and 36.0, sit close to Tokopedia's 39.3; the constructed ratios, by contrast, reflect company-wide monetization.

The relationship is not stable. Revenue grew faster in six of the eight main-sample comparisons, and the median absolute gap between the two growth rates was 38.92 percentage points. In two comparisons, revenue and transaction value moved in opposite directions. Tokopedia's 53.2 percent revenue growth against an 8.9 percent transaction-value decline gives a 62.1-point divergence. After dropping the largest comparison, revenue still grew faster in five of seven, with a median gap of 15.74 points. Every leave-one-out sample still shows revenue growing faster in a majority of comparisons and contains at least one reversal. Adding Grab and Shopee raises the count to ten of twelve, but their four comparisons simply reflect company-wide monetization. Across the issuer-reported series, every Ecosystem Ratio ends below its starting point, although the path is not monotonic in every case. Figure 1 therefore focuses on the stronger dynamic test: the signed growth divergence D across the eight issuer-reported comparisons.

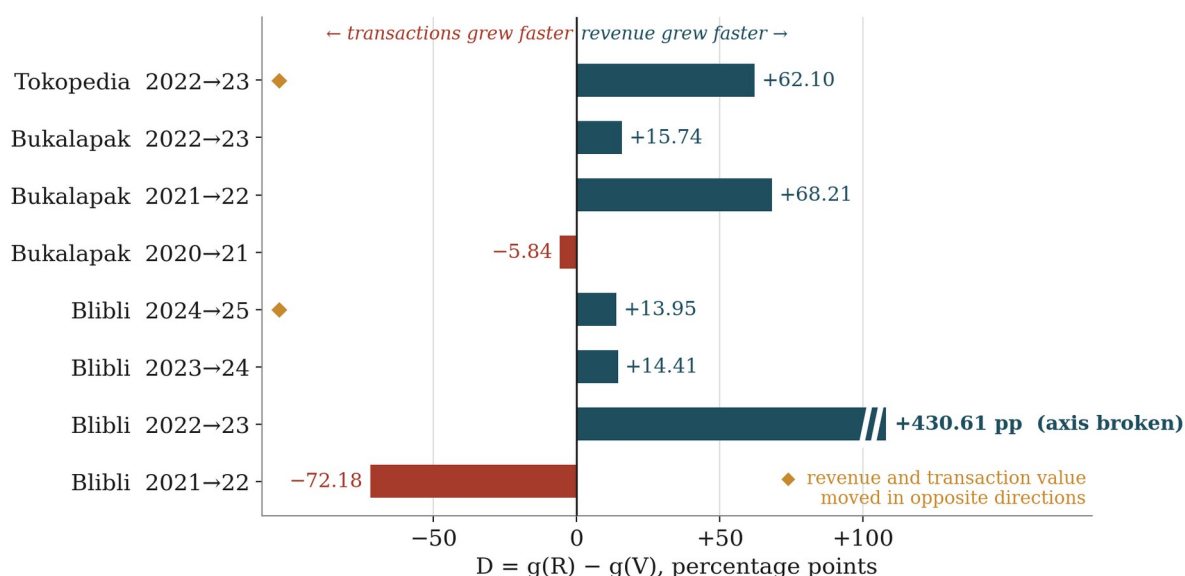


Figure 1. Growth divergence (D) in the issuer-reported main sample, 2020–2025. Positive values indicate faster revenue growth; diamonds mark the two sign reversals. Blibli 2022–2023 is shown with a broken axis.

Tokopedia illustrates the central proxy problem. In 2022–2023, transaction value fell 8.9 percent while third-party net revenue rose 53.2 percent. GoTo's disclosed bridge shows that 60.6 percent of the net-revenue increase reconciles to a lower customer-incentive deduction and 39.4 percent to higher gross revenue. Blibli supplies a second sign reversal in 2024–2025. Across the eight issuer-reported comparisons, revenue grows faster in six, the measures move in opposite directions twice, and median absolute $|D|$ is 38.92 percentage points. These are direct tests of whether revenue preserves the growth story of the commerce it is being asked to represent.

5.2 Evidence Beyond Platform Revenue

National statistics provide a second measurement setting. BPS measures online sales reported by surveyed e-commerce businesses; platform transaction measures use different scopes and definitions, so level differences are treated as scope evidence rather than proof that one source omitted activity. Growth is more directly comparable: between 2023 and 2024, BPS total e-commerce value increased 17.08 percent while its marketplace category increased 1.45 percent and other channels 20.57 percent. On BPS's own definitions, 98.46 percent of the measured increase therefore occurred outside the marketplace category. A component and a wider total can thus support different accounts of the same period without either measure being incorrect.

Growth can be compared more directly. Table 6 summarizes the evidence for H2–H4 and the payment context.

Table 6. Preliminary evidence beyond platform accounts

This table summarizes the preliminary evidence beyond platform accounts for H2–H4 and the payment context. The evidence is drawn from BPS-Statistics Indonesia and Bank Indonesia. Growth figures refer to 2023–2024. The business-count split is arithmetic, and the recordkeeping result is an association.

Question	Preliminary evidence	Interpretation
Where does e-commerce growth come from?	E-commerce value +17.08%; estimated business count +15.31%; sales per business +1.54%. The business count accounts for 90.29% of the increase (70.95% under BPS's other count)	Growth comes mainly from more businesses selling online (H2)
Where does that growth occur?	Marketplace channel +1.45%; other channels +20.57%; 98.46% of the increase is outside the marketplace channel	Most measured growth occurs outside BPS's marketplace category (H4)
Are marketplace use and financial records related?	Complete financial statements: 28.63% of marketplace users versus 12.25% of non-marketplace users	Marketplace participation is associated with stronger recordkeeping (H3)
Do payment indicators show the same growth?	BPS e-commerce +17.08%; electronic-money shopping +30.47%; mobile-banking payments and purchases +82.84%; Quick Response Code Indonesian Standard (QRIS) +186.98%	Context only: payment measures grow 1.8 to 11 times as fast, and also reflect adoption and substitution

BPS also reports widespread use of non-marketplace channels. In 2023, 95.33 percent of e-commerce businesses used instant messaging. In its multiple-response measure, where a business can report several channels, social media accounted for 44.04 percent of transaction value and marketplaces for 32.74 percent; that measure is not comparable with the exclusive 18.2 percent split above. These figures describe the 2023 channel structure rather than the allocation of 2023–2024 growth. The business-count split is arithmetic and the recordkeeping result is an association. Overall, the BPS evidence is consistent with H2, H3 and H4, subject to the definitional differences noted above.

5.3 Feasibility and Scope of the Preliminary Findings

Overall, the preliminary evidence is consistent with each hypothesis, but the measurement argument rests primarily on H1 and H4. H1 is supported by the six of eight issuer-reported comparisons in which revenue outgrew transaction value and by the two sign reversals. H4 is supported by the contrast between platform revenue and transaction growth and by BPS's 17.08 percent total e-commerce growth versus 1.45 percent marketplace growth. H2 and H3 remain secondary national analyses. The central feasibility claim is therefore established: indicator substitution can materially change the measured level, pace and even direction of digital activity.

Outside Indonesia, the same separation appears but its magnitude varies. In 29 clean annual comparisons, revenue grew faster than transaction value in 22 and the measures moved in opposite directions in 4; the median absolute divergence was 7.1 percentage points, compared with 38.92 in the Indonesian issuer-reported sample. The external evidence is used as validation and boundary evidence, not as a representative global estimate. It shows that adequacy is a property of the relationship, scope and period rather than of the source name alone.

PMK 37/2025 provides an administrative example. Under the current DJP schedule, marketplace withholding is deferred through 31 October 2026 and collection is scheduled to begin on 1 November 2026 through designated marketplace operators (Directorate General of Taxes 2026). Those operators cannot yet be mapped directly to BPS's marketplace category; compliance and revenue effects remain outside this study.

6. Why the Problem Matters

Indonesia's digital economy is observed through records that answer different questions. The problem becomes consequential when one record is used as a proxy for another target without testing the relationship between them. Tokopedia shows the sharpest case: recognized revenue implies rapid expansion while transaction value implies contraction. BPS provides a national analogue: its marketplace category grew 1.45 percent in 2023-2024 while total e-commerce grew 17.08 percent. Neither measure is wrong; the inference changes because the economic objects differ. For investors, revenue growth may reflect commerce expansion, monetization, incentives or business mix, with different persistence and valuation implications. For measurement and administration, the coverage of the chosen record must be understood before it represents a broader target. The practical discipline is therefore: define the target, match the scope, test level and movement, reconcile important failures, and state the inference only for the task and period the evidence supports.

7. Limitations and Robustness

This proposal has several limitations that define the framework's boundary. The evidence is descriptive and disclosure-selected: the main sample contains three platforms, 11 platform-years and 8 annual comparisons, so it supports case-based measurement analysis rather than population inference. Coverage is uneven: Tokopedia is observed for two years, Blibli includes online travel, and Bukalapak reports at Group level. Grab and Shopee rely on constructed Indonesian measures and therefore remain separate from the issuer-reported longitudinal result. National, platform and payment indicators also use different definitions; disagreement among them is evidence about measurement, not proof that one source omitted activity. The Tokopedia bridge is an arithmetic reconciliation rather than a causal estimate. Future work can extend matched histories, close the remaining BPS and customer-incentive definitions, and test whether different components of platform growth have different associations with stock returns.

Table 7 pairs each supported conclusion with what it does not establish.

Table 7. Main inference boundaries

This table pairs each conclusion that the evidence supports with what it does not, by itself, establish. Section 7 discusses these inference boundaries.

Evidence supports	Does not by itself establish
W and E describe the level relationship; D tests growth divergence and direction.	Participant profit, taxable income, unpaid tax, tax evasion, or missing GDP.
Issuer-reported pairs show revenue and transaction value moving apart within platforms.	Independent Indonesian ratio evidence from the constructed Grab and Shopee series.
Disclosed revenue components reconcile selected movements in the wedge.	A causal effect of incentives, monetization, or accounting choices.
BPS evidence describes the sources and channels of e-commerce growth and recordkeeping associations.	Causal business entry, productivity effects, or marketplace-induced formalization.
Platform, BPS and payment indicators disagree in level and growth.	Which indicator, if any, contains measurement error, or that payment values measure sales.
PMK 37/2025 draws on designated operators' transaction records.	Its coverage of the digital economy, or a compliance or tax-revenue effect.

One concern is that the results depend on how the sample is defined. Robustness is therefore reported as sensitivity ranges: I vary Shopee's assumed rate, exclude each platform in turn, drop the largest comparison, and restrict the longitudinal result to issuer-reported pairs.

8. Next Steps

The data and empirical framework are assembled. The remaining work is bounded: close the exact source treatment of customer incentives, resolve or disclose the alternative BPS business-count decomposition, complete final source certification and figure wording, incorporate examination feedback, and decide whether the stock-return extension belongs in the final thesis or a subsequent paper. Table 8 sets out that schedule.

Table 8. Planned work toward the final study

This table sets out the planned work toward the final study.

Period	Planned work
Proposal stage	Incorporate examination feedback; close source definitions and final figure wording.
October–November 2026	Resolve BPS sensitivities; complete robustness and source audit; decide and, if feasible, execute the stock-return extension.
Before the final defense	Complete the full manuscript; finalize tables, figures, citations and consistency review.

The study treats indicator choice as an empirical question rather than assuming one record is universally correct. For each task, it defines the economic target and scope, measures the level relationship with W and E, tests dynamic stability with D, reconciles selected failures with disclosed components, and checks whether the relationship survives in other admissible windows or records. The intended output is conditional: which observable is sufficient for which economic conclusion, over which scope and period.

Appendix A. Variables and Data Sources

In Table A1, I report the definition and source of each variable used in the analysis. V and R refer to the same period and business scope, and t, W, E and D are computed from them. Table A2 traces each indicator to the filings and publications behind it.

Table A1. Variable definitions and sources

This table defines each variable used in the analysis and gives its source. V and R refer to the same period and business scope. Growth rates are computed in each issuer's reporting currency, so exchange-rate movements do not enter D. US dollar values are used only for the 2023 cross-section, converted at the World Bank's 2023 official exchange rate of 15,236.88 rupiah per dollar.

Variable	Definition	Source
V	Transaction value processed through platform i in period t: the gross value of the transactions the platform carries, labeled GTV in this proposal. Each issuer's own label, such as GMV or TPV, is retained and checked.	Issuer disclosure, or construction (Table 3)
R	Revenue that platform i recognizes in period t, for the same period and business scope as V.	Issuer disclosure, or construction (Table 3)
t	Monetization rate: the share of transaction value recognized as revenue, $t = R / V$.	Computed
W	Invisible wedge: transaction value outside the platform's recognized revenue, $W = V - R$, in currency units.	Computed
E	Ecosystem Ratio: the wedge per unit of revenue, $E = (V - R) / R = 1/t - 1$. It is constructed for this study as a descriptive transformation of the monetization rate.	Computed
D	Growth divergence: revenue growth minus transaction-value growth, $D = g(R) - g(V)$, where $g(X)$ is the percentage growth of X from one year to the next. A positive D means revenue grew faster than transaction value.	Computed from consecutive years

Table A1. Variable definitions and sources

This table defines each variable used in the analysis and gives its source. V and R refer to the same period and business scope. Growth rates are computed in each issuer's reporting currency, so exchange-rate movements do not enter D. US dollar values are used only for the 2023 cross-section, converted at the World Bank's 2023 official exchange rate of 15,236.88 rupiah per dollar.

Variable	Definition	Source
S	National e-commerce value: the online sales of surveyed businesses, $S = N \times A$.	BPS-Statistics Indonesia
N	Estimated number of e-commerce businesses. BPS publishes two counts for 2023; the count consistent with its stated growth figure is used, and the other is retained as a sensitivity check.	BPS-Statistics Indonesia
A	Implied nominal sales per e-commerce business, $A = S / N$. The change in S is divided arithmetically into the change in N and the change in A.	Computed from BPS-Statistics Indonesia

Table A2. Source lineage by evidence type

This table traces each indicator to the filings and publications behind it. Constructed series are labeled as such wherever they appear. Shopee's Indonesian transaction value is a third-party estimate (Momentum Works 2024), and Indonesian figures for Grab and Shopee are derived from Group ratios (Table 3).

Evidence type	Platform / source	Transaction-value source	Revenue / comparison source
Issuer-reported, Indonesia-aligned	Tokopedia e-commerce	GoTo annual report, e-commerce segment metrics	GoTo annual report, segment note
Issuer-reported, broader scope	Blibli third-party business	Global Digital Niaga prospectus and results releases	Same issuer filings
Issuer-reported, broader scope	Bukalapak Group	Bukalapak annual and sustainability reports	Same issuer filings
Requires construction	Grab Indonesia	Derived from Indonesia revenue and Group monetization rate	Grab Form 20-F
Requires construction	Shopee Indonesia	Momentum Works Southeast Asia e-commerce estimate	Derived from Sea Limited Form 20-F
National statistics	BPS-Statistics Indonesia	E-Commerce Statistics 2023 and 2024; marketplace study	Business-count, channel and recordkeeping evidence
Payment statistics	Bank Indonesia	Payment System Statistics and QRIS reports	Growth-rate comparison only
Exchange rate	World Bank	World Bank official exchange-rate series	Cross-section currency conversion only
External corroboration	eBay, Etsy, Jumia, Mercado Libre, Rakuten, Sea, Shopify, Zalando	Issuer transaction measures	Issuer revenue measures
Regulatory	PMK 37/2025; DJP	Ministry of Finance / Directorate General of Taxes	Institutional architecture only

Every figure used in this proposal traces to a named file and locator in the accompanying data package.

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